LUNARSAFE // AUTONOMOUS MISSION SCAN REPORT

TEAM ROCKET · AEROX CLUB · PARUL UNIVERSITY, VADODARA

Project Head: Darshan Patil | Standard: NASA CLPS / Artemis III | Timestamp: 2026-11-06T03:31:24.991Z

RECOMMENDATION: SAFE [CONFIDENCE: LOW]

Reasoning: Terrain nominal with mean slope of 0.73° (<5°), selected zone max slope of 9.50° (<10°, frame max: 41.21°), 99.2% safe regolith coverage, and minimal zone boulders (0) conforming to NASA CLPS / Artemis III touchdown criteria.

1. TELEMETRY & TERRAIN SAFETY METRICS

Telemetry Parameter	Evaluated Value	Mission Standard / Limit
Scan Record ID	scan-1793935884992-fcb622e0	Unique Telemetry Key
Data Source	NASA PDS / LRO LOLA (Calibrated)	Official NASA Planetary Data System Product
Sensor Instrument	ISRO Chandrayaan-2 TMC-2 (5 m/px GSD)	Orbital Sensor Resolution
Raster Cell Resolution	Actual cell size: 20.0m × 20.0m (400.0 m²)	Target: 24m NASA standard (4 px/cell)
Target Coordinates	No PDS4 label — coordinates unknown.	Selenographic Reference
Mean Surface Slope	0.73°	NASA Threshold: < 5.0° (Nominal)
Maximum Local Slope	41.21°	NASA CLPS Threshold: < 10.0°
Crater Depressions Detected	65	Morphological Hough Extractor
Boulder Obstacles Detected	55	NASA Safety Limit: < 20 rocks
Safe Regolith Coverage	99.2%	NASA Minimum Safe: > 70.0%
Hazard Regolith Area	0.8%	Steep Slopes / Craters / Boulders
Contiguous Safe Clusters	1	4-Connected BFS Search
Primary Landing Site Footprint	6,498,800 m²	Contiguous Obstacle-Free Regolith

2. SHA-256 DETERMINISTIC EVIDENCE CHAIN (TAMPER-EVIDENT AUDIT)

Input Buffer Hash:	`fcb622e0c414539dad7b3ffac8b8b99b7da81ba59407a338b4532cf61ce7fb23`
Calibrated Slope Hash:	`f28a0edce38c66060b2714c43bef3216fbbd0244139e7550a4b6aa3c5d707b7a`
NASA 24m Grid Hash:	`bfd4a138861e05125a8e8399a887069ab140a79fb0793755b8b6823d223c56ab`
Candidate Zones Hash:	`8e3f02a2159b86b09cbae59fc2b1871154f5c0536e872f1d6cf0b6f22f65a58d`
Session Integrity Hash:	`N/A (Offline Session)`

LUNARSAFE // PIPELINE EXECUTION TELEMETRY & DUAL SENSOR OVERLAY

Raw Sensor Orbital Ingestion (Left)

Calibrated NASA 24m Hazard Grid (Right)

3. PIPELINE EXECUTION LOG (SUB-MILLISECOND BENCHMARKS)

Pipeline Module	Status	Duration	Deterministic Output / Details
Panchromatic Validation Filter	DONE	3.0 ms	Chromatic fraction: 0.000% "d 0.100%
Contrast & Brightness Enhancer	DONE	5.4 ms	Histogram dynamic range expanded for 262144
High-Pass Sharpening	DONE	3.3 ms	3x3 kernel convolution applied with zero phase
Sobel Gradient Edge Detector	DONE	16.6 ms	Slope Range: [0.0°, 41.2°], Mean: 0.7°
Grid Feature Density Classifier	DONE	4.3 ms	Detected 65 craters (65 hazardous) and 55
NASA 24m Grid Classifier	DONE	33.5 ms	Total: 16384 cells (Actual cell size: 20.0m × 20.0m,
BFS Contiguous Safe Zone Finder	DONE	26.2 ms	Identified 1 contiguous landing candidates (Top
A* Pathfinding Traversal	DONE	12.5 ms	Calculated path length: 1560.0m across 66
AI-Assisted Mission Validation (Gemini Vision)	IDLE	0.0 ms	Ready for user invocation on upload
AI Narrative Generator (Gemini)	IDLE	0.0 ms	Awaits completed deterministic analysis

INDEPENDENT RESEARCH ATTESTATION

Mission-planning analysis using configurable assumptions (not NASA CLPS requirements). Independent research tool developed for NASA Space Apps 2026.
Project Head: Darshan Patil | Team: Team Rocket | AEROX Club, Parul University